

Pierpaolo Vendittelli

Applied AI Researcher · ML Engineer

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EXPERIENCE

Postdoc Researcher

Dec. 2024 – Present

Prinses Maxima Centre for Pediatric Oncology

Utrecht, The Netherlands

- Developing DL and MIL pipelines for WSI analysis of Wilms tumor (anaplasia grading, clinical outcome prediction) across multiple collaborating hospitals.
- Translating model outputs into clinically interpretable biomarkers in collaboration with pathologists; supervising students and contributing to publications.

PhD Candidate – AI for Digital Pathology

Mar. 2021 – Jun. 2026 (exp.)

Radboud University Medical Center

Nijmegen, The Netherlands

- Developed multi-class U-Net for TSR quantification (PANCAIM, Horizon 2020): 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86, TSR AUC 0.61 on TCGA-PAAD. Extended to multi-tissue segmentation for combining AI biomarkers (TSR, mitosis, stromal density) in PDAC survival analysis. Published in *PLOS ONE* (2024), *SPIE MI* (2022), *COMPAYL* (2025), *European Radiology* (2025), *Lancet Oncology* (2025).
- Teaching Assistant (*Image Analysis*, *Medical Imaging Analysis*, 4 editions); co-supervised Master's theses.

Machine Learning Engineer

Oct. 2020 – Feb. 2021

Everis Italy (NTT DATA)

Rome, Italy

- Built AI chatbots, annotated industrial data for object recognition, and developed live video translation pipelines for clients in banking and automotive.

Research Intern

Feb. 2020 – Sep. 2020

ViCOROB – NeuroImage Computing Group, University of Girona

Girona, Spain

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018; ResU-Net achieved best SSIM/MSE under data-scarce conditions.

SKILLS

Languages Python, C++, C, Bash, SQL

ML / Deep Learning PyTorch, scikit-learn, HuggingFace, Transformers, MONAI, Keras

MLOps & Infrastructure Docker, Git, SLURM / GPU clusters, OpenSlide, W&B

LLM & Agents OpenAI API, Anthropic API, prompt engineering, multi-agent orchestration

PROJECTS & PUBLICATIONS

Weekly AI Paper-Triage Agent

Personal Project

github.com/PierpaoloV/weekly_ai_paper

2024 – Present

Multi-agent pipeline (Anthropic API, Python): weekly arXiv/PubMed scan → LLM summarisation → interest scoring (0–10) → email digest. In active use since 2024.

AI Job-Search Agent

Personal Project

github.com/PierpaoloV/job-agent

2025 – Present

End-to-end Python pipeline: Gmail API ingestion of job-alert emails → structured parsing (LinkedIn, Indeed, Glassdoor) → SQLite deduplication → LLM ranking with prompt caching (Claude Haiku) → Telegram digest. Runs 3×/day via GitHub Actions.

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer

PLOS ONE 2024

May 2024

Multi-class U-Net; 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86; TSR AUC 0.61 for 6-month survival (external TCGA-PAAD cohort).

Automatic tumour segmentation in H&E WSIs of the pancreas

SPIE MI 2022

Apr. 2022

Multi-task U-Net with auxiliary classification head for pancreatic tumour area segmentation; classification head reduced false positives.

EDUCATION

Joint M.Sc. in Medical Imaging and Applications (MAIA)

Sep. 2018 – Sep. 2020

University of Burgundy, University of Cassino, University of Girona

France, Italy, Spain

- Grades: Bien (FR), 102/110 (IT), 8.6/10 (ES) • Erasmus Mundus Consortium Grant

- Grade: 94/110 • Erasmus+ Exchange, Fontys University of Applied Sciences, Eindhoven (2016)